Technical Specification Document

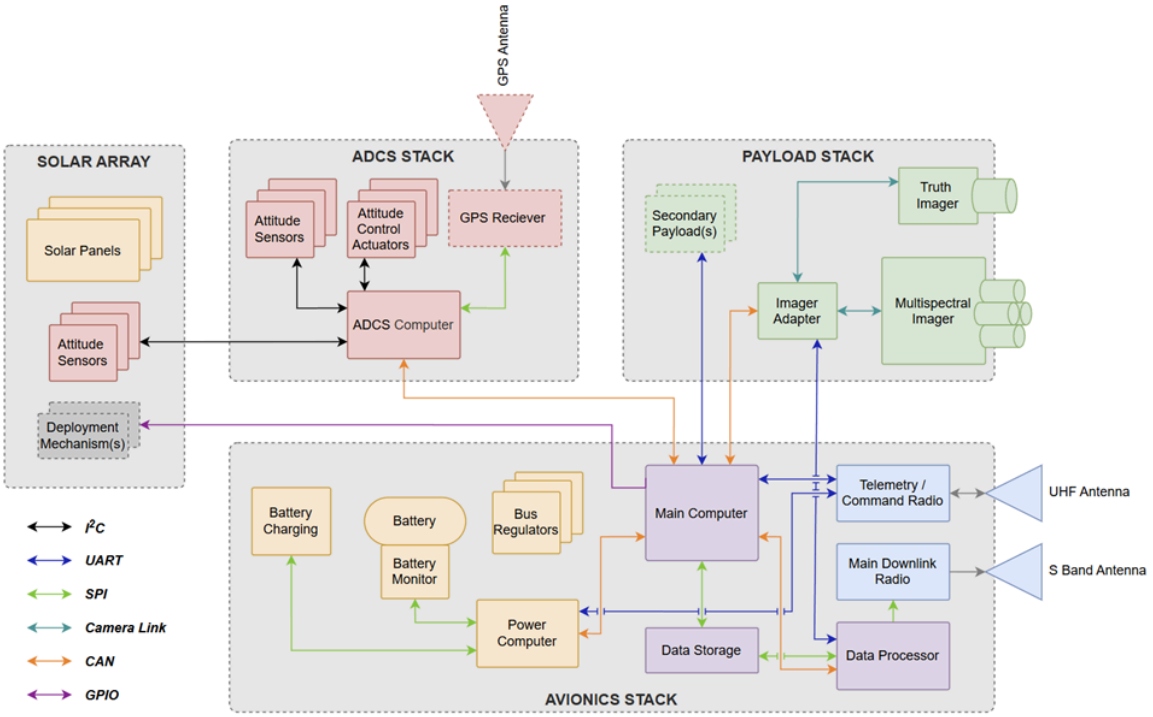
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Version and Change Log

Version	Date	Change Description	Author
0.1	2026-10-02	Created D0 High-Level System Architecture: block diagrams, protocol lines, component responsibility tables, interface specifications, data flow diagrams, and ADRs 001–004 (Assignment 5).	Caden O'Leary, Elmoataz Abdellah, Matthew Goldsberry, Seunghun Lee, Kristen Griessman
0.2	2026-10-08	Created D1 Detailed Design: embedded non-relational data model, O(1) Pass Execution Sequencer algorithm, Build-vs-Reuse decisions, CAN/UART/SPI/GPIO API contracts, and NASA F' + EnduroSat OBC Type I justifications (Assignment 6).	Matthew Goldsberry, Seunghun Lee, Elmoataz Abdellah, Caden O'Leary, Kristen Griessman
0.3	2026-10-14	Created D2 Detailed Design: subsystem cardinality models, Ground Station telecommand table (0x00–0x16), 80 component edge cases (EC-01–EC-80), design rationale, and peer review action items (Assignment 7).	Kristen Griessman, Elmoataz Abdellah, Caden O'Leary, Matthew Goldsberry, Seunghun Lee
1.0	2026-10-28	Compiled unified Technical Specification Document for Assignment 9 mapped directly to the Assignment 9 L0–L4 requirements baseline (L3-OBC-129–176, L4-OBC-185–268) and CON-01–CON-10.	Caden O'Leary, Elmoataz Abdellah, Matthew Goldsberry, Seunghun Lee, Kristen Griessman

Architecture Summary

The HABSat-1 On-Board Computer (OBC) flight software architecture implements a hybrid **Embedded Sensor-Actuator and Layered Pattern** for onboard spacecraft control, paired with a **Client-Server Pattern** for Ground Station command and telemetry links (referencing the **D0 HABSat-1 Data and Power Lines Block Diagram** and **D0 HABSat-1 Protocol Lines Diagram** from Assignment 5).



HABSat-1 Protocol Lines

Defines the hardware communication protocols between components in HABSat-1.

Major System Components

As defined in the D0 architecture and refined in D1/D2, the avionics and subsystem topology comprises twelve interacting nodes orchestrated by the Main Computer:

1. **Main Computer (OBC):** Hosted on the EnduroSat OBC Type I board (STM32F407VG ARM Cortex-M4 MCU at 168–180 MHz, 1 MB Flash, 192+4 KB SRAM). Executes FreeRTOS (L4-OBC-247) and NASA F' components for the 10-mode Finite State Machine (L4-OBC-209), telecommand decryption/validation (L4-OBC-185–196), 60-second telemetry polling (L4-OBC-220), 16-pass imaging sequencer (L4-OBC-235–243), and triple-watchdog FDIR (L4-OBC-250).
2. **Data Processor (DP):** Dedicated secondary coprocessor (RP2350, dual Cortex-M33/Hazard3 cores with 520 KB SRAM, added per **ADR-001** and L3-OBC-167) that ingests raw image blocks from the Imager Adapter over UART, computes the Chloropyhll Index (L4-OBC-244) and lossless compression (L4-OBC-223), writes processed files to Data Storage, and streams high-rate imagery directly to the Main Downlink Radio (**ADR-002**, L3-OBC-133).
3. **Data Storage (DS):** Dual-tier non-volatile memory over SPI comprising radiation-tolerant **MRAM** (L3-OBC-145, L4-OBC-212, L4-OBC-264, added per Peer Review Action #3 for 30-second state flags, boot counters, inhibit timers, and parameter tables) and **NAND Flash** (L3-OBC-144, L4-OBC-224–230 for CRC-verified multispectral images and FIFO telemetry logs).
4. **ADCS Computer & Stack:** Dedicated attitude controller interfacing with Attitude Sensors (sun sensors, magnetometer, gyroscopes over I2C), Attitude Control Actuators (reaction wheels and magnetorquers), and the GPS Receiver/Antenna (SPI + 1PPS). Computes onboard orbit propagation from forwarded TLEs (L4-OBC-234) and executes closed-loop pointing (L3-OBC-153).
5. **Power Computer (EPS) & Stack:** Manages Solar Panels, Battery Charging (MPPT), Battery Monitor, and Bus Regulators. Switches subsystem power rails (L4-OBC-210, L4-OBC-261), monitors the Main Computer watchdog heartbeat (L4-OBC-258), and maintains a backup UART link to the UHF Telemetry Radio for Main Computer failover (**ADR-004**).
6. **Imager Adapter & Optical Payloads:** Controls the **Multispectral Imager** (620, 650, 680, 708 nm) and **Truth Imager** via Camera Link, receives warmup/capture commands from the Main Computer over CAN (**ADR-003**, L4-OBC-238–239), and streams raw image blocks to the Data Processor over UART (L4-OBC-245).
7. **Secondary Payload:** External experimental payload (MyRadar candidate) connected to the Main Computer via a dynamically gated UART interface.
8. **Telemetry / Command Radio (UHF) & UHF Antenna:** Half-duplex UART transceiver handling encrypted Ground Station telecommand uplinks (L3-OBC-129), 49-second unencrypted housekeeping beacons (L4-OBC-203–205), on-demand engineering telemetry downlinks (L3-OBC-135), backup UHF image thumbnails (L3-OBC-134, L4-OBC-201), and amateur radio digipeater frames (L3-OBC-155, L4-OBC-207–208).

9. **Main Downlink Radio (S-Band) & S-Band Antenna:** High-data-rate SPI transmitter connected directly to the Data Processor (**ADR-002**) to downlink segmented multispectral and CI imagery with flow control (L3-OBC-133, L4-OBC-199–200).
10. **Deployment Mechanisms:** Discrete GPIO-controlled burn-wire release circuits (PIN_DEP_SOLAR_ACT, PIN_DEP_CAMERA_LENS, PIN_DEP_ANTENNAS) with 2.5 A current limiting (L4-OBC-214) and 100 ms microswitch debouncing (L4-OBC-215).
11. **Ground Station (GS):** Earth terminal running the Python F' Ground Data System (GDS), an SQLite telemetry/image database (Task 022), and a local JSON API / React monitoring dashboard (Tasks 023–024).

Deployment Model & Data Flow

Flight software is compiled as a statically linked C++17 firmware image deployed onto the Main Computer's internal NOR Flash (with a write-protected golden fallback image in ROM/MRAM per L3-OBC-166 and L4-OBC-251) and a companion firmware image on the RP2350 Data Processor. At runtime, data flows across four hardware bus topologies as shown in the D0 Protocol Lines Diagram:

- **Control & Housekeeping Flow (CAN 2.0B Bus):** The Main Computer exchanges fixed 12-byte CAN frames with the ADCS Computer, Power Computer, Imager Adapter, and Data Processor. Every 60 seconds, the Main Computer polls health telemetry from all nodes (L3-OBC-138, L4-OBC-220); asynchronous fault frames (MSG_PWR_LOW_BATTERY_ALERT, MSG_IMG_HARDWARE_FAULT, MSG_ADCS_POINTING_TIMEOUT) preempt background tasks within 100 ms (L3-OBC-139, L4-OBC-222).
- **High-Rate Science Data Flow (Camera Link -> UART -> SPI -> S-Band):** Prior to an imaging pass, the Main Computer verifies SoC >= 20% and GSD <= 90 m (L4-OBC-237), then commands ADCS target tracking and Imager warmup over CAN (L4-OBC-238). Once MSG_ADCS_TARGET_ACQUIRED and MSG_IMG_WARMUP_READY assert, the Main Computer triggers burst capture (L4-OBC-239) and records synchronized ECEF/LVLH georeference vectors (L4-OBC-260). Raw frames stream from the Imager Adapter over UART to the Data Processor (with 3x checksum retry per L4-OBC-245) for Chloropyhll Index calculation (L4-OBC-244) and lossless compression (L4-OBC-223), and are written over SPI to Data Storage with 32-bit CRCs (L4-OBC-224). Upon Ground Station command (0x11), the Data Processor streams segmented frames over SPI directly to the S-Band Main Downlink Radio (**ADR-002**, L4-OBC-199–200).
- **Telecommand & Beacon Flow (UHF RF <-> UART <-> Main Computer):** Encrypted Ground Station telecommands (0x00–0x16) arrive at the UHF Radio and pass over UART to the Main Computer, where Radio_SecurityHandler decrypts the ASCON-AEAD128 payload (L4-OBC-185), verifies the 16-byte authentication tag (L4-OBC-186) and sequence counter (L4-OBC-187), validates syntax/bounds/mode (L4-OBC-190–192), and logs/transmits an ACK or NACK (L4-OBC-194–196).

Component Interfaces and Data Formats

Wire Framing & Encryption Schemas

Internal CAN Bus Frame Schema (Fixed 12 Bytes — Main Computer <-> ADCS, EPS, Imager Adapter, Data Processor)

Byte Offset	Field Name	Size	Type / Value	Description
Byte 0	CAN_STR	1 Byte	uint8_t (0xAA)	Start-of-frame delimiter byte.
Byte 1	CAN_ID	1 Byte	uint8_t (0x01–0x73)	Command, ACK, Telemetry, or Async Alert opcode identifier.
Bytes 2–9	CAN_PAYLOAD	8 Bytes	uint64_t / float32[2]	Fixed 64-bit command arguments, sensor bitmasks, or IEEE 754 float telemetry values.
Byte 10	CAN_CRC	1 Byte	uint8_t (CRC-8-CCITT)	Frame integrity checksum calculated over Bytes 0–9.
Byte 11	CAN_ETX	1 Byte	uint8_t (0x55)	End-of-transmission delimiter byte.

Internal UART Frame Schema (10 + N Bytes — Main Computer <-> UHF Radio, Secondary Payload; Imager Adapter <-> Data Processor)

Byte Offset	Field Name	Size	Type / Value	Description
Bytes 0–1	PREAMBLE	2 Bytes	uint16_t (0x7E7E)	Synchronization preamble bytes.
Bytes 2–5	SYNC_WORD	4 Bytes	uint32_t (0x43554245)	Frame alignment sync word.
Byte 6	LENGTH	1 Byte	uint8_t (N <= 245)	Dynamic payload byte length N.
Byte 7	NODE_FRAME_ID	1 Byte	uint8_t (0xA1–0xD1 / 0x00–0x16)	Target subsystem address/node and frame command/telemetry ID.
Bytes 8 to 7+N	PAYLOAD	N Bytes	uint8_t[N]	Unencrypted beacon/digipeater payload OR ASCON-AEAD128 encrypted block.
Bytes 8+N to 9+N	CRC16	2 Bytes	uint16_t (CRC-16-CCITT)	Error-detection CRC across header and payload (L4-OBC-190).

ASCON-AEAD128 Encrypted Radio Payload Schema (Up to 255 Bytes Total inside UART/RF Frame)

Field	Size	Description
IV (Initialization Vector)	16 Bytes (128-bit)	Unique per-packet nonce comprising a 32-bit epoch timestamp + 32-bit monotonic replay counter (L4-OBC-187) + 64-bit mission salt.

Field	Size	Description
Header Data (Associated Data)	4 Bytes	Unencrypted Cleartext Command/Packet ID (uint8_t), Sequence Counter (uint16_t), and Target Mode Mask (uint8_t) authenticated by AEAD.
Ciphertext	Up to 219 Bytes	ASCON-AEAD128 encrypted command parameters or private telemetry payload (Combined Header + Ciphertext <= 223 bytes).
Authentication Tag	16 Bytes (128-bit)	Cryptographic MAC tag verified prior to plaintext parsing (L4-OBC-186); invalid tag triggers immediate drop and increments security counter (L4-OBC-189, EC-35).

In-Memory C++ Data Structures & Ground Station SQLite / JSON Schemas

- **Onboard Pass Schedule Record (PassRecord C++ struct, 24 bytes static — L4-OBC-235):** struct PassRecord { uint16_t seq_id; uint32_t acq_start_epoch; uint32_t acq_end_epoch; float target_lat; float target_lon; float target_alt; uint8_t priority; uint8_t status; }; (Statically allocated array PassRecord pass_queue[16] on Main Computer per L4-OBC-248).

Component Boundary Matrix (Inputs, Outputs, Formats, Protocol, Error Behavior)

Component Boundary	Inputs (to Target)	Outputs (from Target)	Data Format	Protocol	Error & Edge Case Behavior (D2 Reference)
Main Computer <-> ADCS Computer	Mode commands (0x02), TLE chunks (0x03), Target Lat/Lon/Alt (0x04–0x06), Schedule poll (0x07), Stop Track (0x08).	Command ACK (0x40), Detumble done (0x41), Target lock (0x42), Pass schedule (0x43), Pointing timeout (0x44), Status (0x45).	12-Byte Fixed CAN Frame (uint64_t / IEEE 754 float32).	CAN 2.0B (500 kbps)	CRC-8 check + up to 3 automatic retries (EC-03, L4-OBC-252). On CAN bus-off, reset the controller up to 3x (EC-68). If MSG_ADCS_POINTING_TIMEOUT (0x44) asserts, abort imaging pass and command Minimal Drag (L4-OBC-241).
Main Computer <-> Power Computer (EPS)	Get Vitals (0x20), Rail On/Off (0x21/0x22), Get Charge (0x23), Watchdog Heartbeat (L4-OBC-258).	Rail ACK (0x50), Low Battery Alert (0x51), Charge Recovered (0x52), Vitals (0x53), Deploy Confirm (0x54).	12-Byte Fixed CAN Frame (uint16_t mV, mA, deg C, % SoC).	CAN 2.0B (500 kbps)	Up to 3 read retries before falling back to safe power estimate (EC-23). Stagger simultaneous rail enabled by 50 ms (EC-22). If MSG_PWR_LOW_BATTERY_ALERT (0x51, SoC < 15%) asserts, shed payload rails within 100 ms and enter Low-Power/Safe Mode (EC-21, L4-OBC-222, L4-OBC-254).
Main Computer <-> Imager Adapter	Init Warmup (0x30), Trigger Burst (0x31), Get	ACK (0x60), Warmup Ready (0x61), Capture	12-Byte Fixed CAN Frame (Camera ID,	CAN 2.0B (500 kbps)	If warmup > 60 s or MSG_IMG_HARDWARE_FAULT (0x63) asserts, cut camera

Component Boundary	Inputs (to Target)	Outputs (from Target)	Data Format	Protocol	Error & Edge Case Behavior (D2 Reference)
	Status (0x32), Purge Buffer (0x33).	Complete (0x62), Hardware Fault (0x63).	Frame Count, Exposure ms).		power (0x22) and abort pass (EC-52, EC-60, L4-OBC-241).
Imager Adapter <-> Data Processor	Raw multi-band and truth image frame blocks with line-sync markers and block sequence IDs.	Block ACK / NACK retry request.	Dynamic N-byte UART Frame (Preamble, ID, Len16, Payload, CRC16).	High-Speed UART (3.0 Mbps DMA)	Re-synchronize missing line-sync up to 3x (EC-51). Data Processor verifies block CRC16 and retries corrupted blocks up to 3x before discarding (EC-75, L4-OBC-245).
Main Computer <-> Data Processor	Start Processing (0x10), Abort (0x11), Trigger S-Band Downlink (0x12).	ACK (0x70), Processing Done (0x72), Downlink Status (0x73).	12-Byte Fixed CAN Frame (Pipeline ID, Image File ID, Byte count).	CAN 2.0B (500 kbps)	Arithmetic guard catches divide-by-zero during CI calculation and falls back to storing raw uncompressed image (EC-73, L4-OBC-246). Main Computer enforces execution timeout and disables DP rail (0x22) if hung (L4-OBC-262).
Main Computer & Data Processor <-> Data Storage	Read/Write State Flags (0x0010), Write Log (0x0100), Check/Write Image Index (0x0200), Image chunks + CRC32.	3-byte mission flag bitmask, Stored/Pending status, File blocks, Free capacity.	4-Wire SPI Register + Block Transaction with 32-bit CRC (L4-OBC-224).	SPI (20 MHz) with FreeRTOS Mutex	Mutex locking serializes concurrent Main Computer and Data Processor writes (EC-70, L4-OBC-253). Bad flash blocks remap automatically (EC-71, L4-OBC-227). If SPI bus drops, buffer logs in SRAM ring buffer (EC-72).
Data Processor <-> Main Downlink Radio (S-Band)	Segmented image transfer frames with sequence numbers and CRC16 (L4-OBC-199).	DMA buffer status feedback (L4-OBC-200) and synthesizer lock status.	SPI DMA Frame Stream.	SPI (42 MHz DMA)	On the DMA buffer underrun, insert idle frames (EC-38). If PA temp > 85 deg C, throttle data rate by 50% until < 70 deg C (EC-37). On truncated block, re-fetch from storage (EC-39).
Main Computer <-> Telemetry/Command Radio (UHF)	Beacon TX (0xA1), Telemetry/Thumbnail Downlink (0xA2), Digipeat Stage (0xA3).	Uplink Telecommands (0x00–0x16) and Amateur Repeater frames (0x02).	10+N Byte UART Frame (with ASCON-AEAD128 payload for non-amateur links).	UART (115.2 kbps, Dynamically Gated)	Uplink reception preempts outgoing beacon/digipeater on half-duplex channel (EC-34, L4-OBC-206). Oversized packets (>255 B) drop with error (EC-33). Failed ASCON-AEAD128 tag drops immediately (EC-35, L4-OBC-189).
Main Computer <-> Deployment Mechanisms	High-to-Low / Low-to-High discrete actuation pulse (PIN_DEP_SOLA	Microswitch deployment sense feedback (0 or 1).	Discrete GPIO Digital Logic (3.3 V CMOS) + PWM current limit.	GPIO	Enforces 30-min post-deployment inhibit timer (EC-78, L4-OBC-213). PWM clamps burn-wire current to <= 2.5 A (EC-77, L4-OBC-214).

Component Boundary	Inputs (to Target)	Outputs (from Target)	Data Format	Protocol	Error & Edge Case Behavior (D2 Reference)
	R_ACT, PIN_DEP_CAMERA_LENS, PIN_DEP_ANTENNAS).				Debounces microswitch over 100 ms and retries up to 2x (EC-43, EC-79, L4-OBC-215).
Main Computer <-> Secondary Payload	Ping (0xB1), Start Task (0xB2), Read Data (0xB3).	Frame ACK/NACK (0xD0), Experiment Data Packet (0xD1).	10+N Byte UART Frame.	UART (115.2 kbps)	Rejects corrupted command scripts (EC-63). Buffers data locally during high-priority tasks (EC-62). Saves state to NVM prior to load-shedding power cut (EC-64).

API Endpoint & Command Tables

Ground Station <-> Satellite Telecommand Uplink API (UHF Radio -> Main Computer)

Cmd ID	Command Name	Direction	Encrypt	Auth Modes	Request Payload & Behavior	Response / Error Behavior
0x00	Initiate Main Mission	GS -> OBC	TRUE	Heartbeat / Checkout	0-byte arg. Sets MISSION_COMMISSIONED_FLAG = TRUE in SPI MRAM (0x0010) and transitions FSM to Standby Mode (L4-OBC-209).	Queues RADIO_SEND_DOWNLINK (0xA2) ACK (L4-OBC-194); NACK 0xE001 if inhibit active (L4-OBC-195).
0x01	Transmit Telemetry	GS -> OBC	TRUE	All Modes	uint8_t telemetry mask (Public/Private). Compiles current subsystem health and historical logs for UHF downlink (0xA2, L3-OBC-135, L4-OBC-202).	Streams telemetry frames (DL-01–DL-10); retries up to 3x on link loss (EC-36).
0x02	Repeat Message	Amateur GS -> OBC	FALSE	Repeater / Standby	Dynamic amateur callsign + message payload (<= 200 B). Stages in 64-kb ring buffer (L4-OBC-207) for UHF digipeater rebroadcast (0xA3, L3-OBC-155).	Rebroadcasts (DL-12) within duty cycle limit (L4-OBC-208); drops if in Low-Power/Imaging/Inhibit.
0x03	Exit Safe Mode	GS -> OBC	TRUE	Safe Mode	uint16_t fault-clear mask. Clears latched fault flags in MRAM and transitions FSM from Safe Mode to Checkout Mode (L3-OBC-162, L4-OBC-255).	Returns ACK (0xA2); NACK 0xE002 if battery SoC < 40% (EC-21).
0x04	Enter Low-Power Mode	GS -> OBC	TRUE	All Nominal Modes	0-byte arg. Sheds non-essential payload/DP rails (0x22), commands ADCS to Sun-Pointing (0x02), and suspends beacons (L4-OBC-205).	Returns ACK (0xA2) before entering Low-Power Mode.

Cmd ID	Command Name	Direction	Encrypt	Auth Modes	Request Payload & Behavior	Response / Error Behavior
0x05	Power Cycle Systems	GS -> OBC	TRUE	All Modes (2-Step Confirm)	uint8_t rail bitmask (0x00 Imagers, 0x01 DP, 0x02 S-Band, 0x03 ADCS). Commands EPS (0x22 then 0x21 with 50 ms stagger).	Requires secondary confirmation (L4-OBC-193); returns MSG_PWR_ACK.
0x06	Reboot Flight Computer	GS -> OBC	TRUE	All Modes (2-Step Confirm)	uint8_t boot target (0x00 Primary Flash, 0x01 Golden ROM). Saves state to MRAM (L4-OBC-212) and triggers controlled reset.	Requires secondary confirmation (L4-OBC-193); transmits ACK prior to reboot.
0x07	Enter RF Inhibit	GS -> OBC	TRUE	All Modes (2-Step Confirm)	uint32_t optional duration timer. Disables UHF and S-Band RF transmitter stages (L3-OBC-171, L4-OBC-205).	Requires secondary confirmation (L4-OBC-193); logs state in MRAM and silences RF transmitters.
0x08	Exit RF Inhibit	GS -> OBC	TRUE	All Modes	0-byte arg. Restores RF transmitter enable flags and resumes nominal passes and 49-s housekeeping beacons (L3-OBC-171, L4-OBC-204).	Immediately transmits ACK (0xA2) and resumes the beacon timer.
0x09	Enter Emergency Mode	GS -> OBC	TRUE	All Modes (2-Step Confirm)	0-byte arg. Cuts all high-draw rails, locks ADCS to Sun-Pointing (0x02), and switches to minimal carrier beaconing (L4-OBC-209).	Requires secondary confirmation (L4-OBC-193); logs emergency entry in MRAM (0x0010).
0x0A	Exit Emergency Mode	GS -> OBC	TRUE	Emergency Mode	uint16_t verification token. Returns spacecraft to Checkout or Standby Mode after ground verifies bus stability.	Returns ACK (0xA2); NACK 0xE003 if critical bus faults remain latched.
0x0B	Send Errors	GS -> OBC	TRUE	All Modes	uint16_t log count. Reads historical fault logs, bus reset counters, and watchdog flags from SPI Storage (0x0100, L3-OBC-136).	Downlinks encrypted DL-11 Error Log packets over UHF (0xA2).
0x0F	Downlink Sec Payload Pkt	GS -> OBC	TRUE	Standby / Downlink	uint16_t packet index. Polls Secondary Payload (0xB3) or storage for a single experiment packet and downlinks over UHF.	Returns SEC_PAY_DATA_PACKET (0xD1) wrapped in UHF downlink frame.
0x10	Downlink Sec Payload Range	GS -> OBC	TRUE	Standby / Downlink	uint16_t start_idx, uint16_t end_idx. Streams sequential block of Secondary Payload packets over UHF (L4-OBC-199).	Returns segmented transfer frames; NACK if range invalid (L4-OBC-191).
0x11	Send Specified Images	GS -> OBC	TRUE	Standby / Downlink	uint16_t image_id, uint8_t band_mask. Enables Data Processor & S-Band rails (0x21) and issues	Streams encrypted full-res/CI image (DL-05) over S-Band; reports 0x73 status.

Cmd ID	Command Name	Direction	Encrypt	Auth Modes	Request Payload & Behavior	Response / Error Behavior
					CMD_DP_TRIGGER_DOWNLINK (0x12, L3-OBC-133).	
0x12	Uplink Orbit TLE Set	GS -> OBC	TRUE	Standby / Checkout	uint8_t buf_id + 138-byte NORAD Two-Line Element string. Validates checksum and forwards to ADCS via CMD_ADCS_LOAD_TLE (0x03, L3-OBC-151, L4-OBC-234).	Forwards to ADCS and polls CMD_ADCS_REQ_SCHEDULE (0x07); returns ACK.
0x13	Send Image via UHF	GS -> OBC	TRUE	Standby / Downlink	uint16_t image_id. Retrieves compressed thumbnail and ECEF/LVLH georeference header from SPI Storage and routes over UHF (0xA2, L3-OBC-134, L4-OBC-201).	Streams encrypted DL-06 thumbnail packets over UHF.
0x14	Purge Stored Image	GS -> OBC	TRUE	Standby / Downlink (2-Step Confirm)	uint16_t image_id. Deletes ground-verified image file in SPI Storage and issues CMD_IMG_PURGE_BUFFER (0x33, L3-OBC-147, L4-OBC-225).	Requires secondary confirmation (L4-OBC-193); frees storage blocks and returns ACK.
0x15	Send Target Coordinates	GS -> OBC	TRUE	Standby / Checkout	float32 lat, float32 lon, float32 alt (12 bytes IEEE 754). Forwards to ADCS (0x04–0x06, L4-OBC-234) and requests updated schedule (0x07).	Rejects if Lat outside [-90, 90] or Lon outside [-180, 180] (L4-OBC-191).
0x16	Sync Time	GS -> OBC	TRUE	All Modes	uint32_t unix_epoch_sec. Disciplines spacecraft epoch clock when ADCS GPS 1PPS is unavailable (L3-OBC-143, L4-OBC-219).	Updates RTC without disturbing monotonic pass timeout counter (L4-OBC-243).

Internal Subsystem Command, Telemetry & Register API Endpoints (Main Computer <-> Subsystems)

Bus / Protocol	Endpoint ID / Pin	Symbol Name	Direction	Type	Payload Schema / Register Format	Functional & Error Behavior
CAN (ADCS)	0x01 / 0x45	CMD_ADCS_GET_STATUS / MSG_ADCS_STATUS_DATA	OBC <-> ADCS	Telemetry Poll	Req: 0x00000000 -> Resp: 64-bit sensor validity flags & body rates.	Polled every 60 s (L4-OBC-220); timeout > 50 ms retries 3x (L4-OBC-252).
CAN (ADCS)	0x02 / 0x40	CMD_ADCS_SET_MODE /	OBC <-> ADCS	Command Req	0x01 Detumble, 0x02 Sun-Point, 0x03 Min Drag,	Synchronous 0x40 ACK (0x00 = OK,

Bus / Protocol	Endpoint ID / Pin	Symbol Name	Direction	Type	Payload Schema / Register Format	Functional & Error Behavior
		MSG_ADCS_ACK			0x04 Track, 0x05 GS Point.	uint16_t err code) (L3-OBC-153).
CAN (ADCS)	0x03–0x06	CMD_ADCS_LOAD_TLE / SET_TARGET_LAT/LON/ALT	OBC -> ADCS	Command Req	0x03: uint8_t buf_id, uint56_t chunk; 0x04–0x06: float32 coordinate.	Transfers uplinked TLEs (0x12) and lake target coords (0x15, L4-OBC-234).
CAN (ADCS)	0x07 / 0x43	CMD_ADCS_REQ_SCHEDULE / MSG_ADCS_SCHEDULE_PACKET	OBC <-> ADCS	Telemetry Poll	Resp: uint8_t pass_id, uint24_t start_ts, uint24_t end_ts, uint8_t prio.	Populates 16-slot pass_queue (L4-OBC-235); resolves overlaps (L4-OBC-236).
CAN (ADCS)	0x08	CMD_ADCS_STOP_TRACK	OBC -> ADCS	Command Req	0x00000000 (64-bit zero payload).	Ends closed-loop tracking upon capture completion (L4-OBC-240) or pass abort (L4-OBC-241).
CAN (ADCS)	0x41 / 0x42 / 0x44	MSG_ADCS_DETUMBLE_COMPLETE / TARGET_ACQUIRED / POINTING_TIMEOUT	ADCS -> OBC	Async Event / Alert	64-bit status/error payload.	0x41 sets DETUMBLE flag; 0x42 gates capture (L4-OBC-239); 0x44 aborts pass (L4-OBC-241).
CAN (EPS)	0x20 / 0x53	CMD_PWR_GET_VITALS / MSG_PWR_VITALS_DATA	OBC <-> EPS	Telemetry Poll	Resp: uint16_t mV, int16_t mA, int16_t temp_c, uint16_t soc_pct.	Polled every 60 s (L4-OBC-220) and strobes EPS watchdog (L4-OBC-258).
CAN (EPS)	0x21 / 0x22 / 0x50	CMD_PWR_RAIL_ON / RAIL_OFF / MSG_PWR_ACK	OBC <-> EPS	Command Req	Rail ID: 0x00 Imagers, 0x01 Data Processor, 0x02 S-Band Radio.	Staggered 50 ms switching (EC-22); verified via 0x50 ACK (L4-OBC-210, L4-OBC-261).

Bus / Protocol	Endpoint ID / Pin	Symbol Name	Direction	Type	Payload Schema / Register Format	Functional & Error Behavior
CAN (EPS)	0x23 / 0x54	CMD_PWR_GET_CHARGE_STATE / MSG_PWR_DEPLOY_CONFIRM	OBC <-> EPS	Telemetry Poll	Queries solar array MPPT current generation and deployment verify.	Confirms solar array deployment after PIN_DEP_SOLAR_ACT pulse (L3-OBC-170).
CAN (EPS)	0x51 / 0x52	MSG_PWR_LOW_BATTERY_ALERT / CHARGE_RECOVERED	EPS -> OBC	Async Alert	uint8_t soc_pct (0x51 triggers at SoC < 15%; 0x52 at SoC >= 40%).	0x51 preempts tasks in <100 ms (L4-OBC-222), sheds rails, enters Safe/Low-Power Mode (L4-OBC-254).
CAN (Imager)	0x30 / 0x61	CMD_IMG_INIT_WARMUP / MSG_IMG_WARMUP_READY	OBC <-> Imager	Command / Event	0x01 Truth, 0x02 Multispectral; 0x61 returns sensor float32 temp.	Aborts pass if warmup exceeds 60 monotonic seconds (L4-OBC-238, L4-OBC-241, L4-OBC-243).
CAN (Imager)	0x31 / 0x62 / 0x63	CMD_IMG_TRIGGER_BURST / CAPTURE_COMPLETE / HARDWARE_FAULT	OBC <-> Imager	Command / Event	uint8_t cam_id, uint8_t frames, uint16_t exp_ms (capped <= 10,000 ms, EC-53).	0x62 triggers DP pipeline (L4-OBC-244); 0x63 (0x01 Timeout, 0x02 Link) aborts pass (L4-OBC-241).
CAN (Imager)	0x32 / 0x33 / 0x60	CMD_IMG_GET_STATUS / PURGE_BUFFER / MSG_IMG_ACK	OBC <-> Imager	Command / Poll	0x33: uint16_t target_image_id.	0x33 invoked only after Ground Station 0x14 purge command (L4-OBC-225).
CAN (Data Proc)	0x10 / 0x11 / 0x72	CMD_DP_START_PROCESSING / ABORT / MSG_DP_PROCESSING_DONE	OBC <-> DP	Command / Event	0x10: Pipeline 0x01 (CI Conversion); 0x11: Reason (0x01 Low Pwr, 0x02 Preempt).	0x72 returns generated CI image count (L4-OBC-244); guarded by execution

Bus / Protocol	Endpoint ID / Pin	Symbol Name	Direction	Type	Payload Schema / Register Format	Functional & Error Behavior
						timeout (L4-OBC-262).
CAN (Data Proc)	0x12 / 0x70 / 0x73	CMD_DP_TRIGGER_DOWNLINK / MSG_DP_ACK / DOWNLINK_STATUS	OBC <-> DP	Command / Resp	0x12: uint16_t file_id; 0x73: uint24_t bytes_tx, uint8_t status.	Streams stored imagery over SPI to S-Band radio (ADR-002, L3-OBC-133, L4-OBC-200).
UART (UHF)	0xA1 / 0xA2 / 0xA3	RADIO_BEACON_TX / SEND_DOWNLINK / DIGIPEAT_STAGING	OBC -> Radio	Command Req	10+N byte UART frame (up to 255 B).	Half-duplex arbiter (EC-34, L4-OBC-206) prioritizes incoming telecommand reception.
SPI (Storage)	0x0010	DS_READ_FLAGS / DS_WRITE_FLAG	OBC <-> DS	SPI Register	16-bit reg 0x0010, 3-byte bitmask (SOLAR_DEPLOY, DETUMBLE, COMMISSIONED).	Stored in radiation-tolerant SPI MRAM every 30 s (L3-OBC-145, L4-OBC-212).
SPI (Storage)	0x0100 / 0x0200	DS_WRITE_LOG / DS_CHECK_IMAGE_INDEX	OBC <-> DS	SPI Command	0x0100: uint8_t len + log struct; 0x0200: returns 0x01 Stored / 0x00 Pending.	Serialized via FreeRTOS mutex (EC-70, L4-OBC-253); verifies DP image offload before cutting power (L4-OBC-211).
GPIO (Deploy)	PIN_DEP_*	PIN_DEP_SOLAR_ACT, CAMERA_LENS, ANTENNAS	OBC -> Deploy	Discrete GPIO	Pulse High (1 for 5 s burn cycle, then 0); PWM current limit <= 2.5 A (L4-OBC-214).	Inhibited during first 30 min uptime (L4-OBC-213); debounced 100 ms with 2 retries (L4-OBC-215).
UART (Sec Pay)	0xB1–0xB3 / 0xD0–0xD1	SEC_PAY_PING / START_TASK / READ_DATA /	OBC <-> Payload	Command / Resp	10+N byte UART frame.	Verifies POST responsiveness (0xB1) and polls secondary

Bus / Protocol	Endpoint ID / Pin	Symbol Name	Direction	Type	Payload Schema / Register Format	Functional & Error Behavior
		ACK / DATA_PACKET				science data (0xB3).

Technology Stack

Layer	Technology Choice	License / Version	Justification
Primary Flight Computer Hardware	**EnduroSat OBC Type I Board** (STM32F407VG ARM Cortex-M4F MCU, 168–180 MHz, 1 MB Flash, 192+4 KB SRAM, dual accelerometers/magnetometers, PC/104)	Existing Club Hardware (Space-Qualified)	**Satisfies CON-01 ($0 hardware cost — already owned by CubeCats), L2-SYS-050, L3-OBC-157, L3-OBC-164, L3-OBC-165, L4-OBC-217, and L4-OBC-259:** Purpose-built for ultra-low power CubeSat operation (104–123 mA at 180 MHz in Standby), qualified across -40 deg C to +60 deg C in a 95x95x25 mm PC/104 footprint, with native CAN 2.0B, SPI, UART, I2C, board-level inertial/magnetic sensors (L4-OBC-259), and hardware watchdog (L4-OBC-250).
Secondary Payload Data Processor Hardware	**RP2350 Microcontroller** (Dual ARM Cortex-M33 / Hazard3 cores, 520 KB SRAM, hardware glitch detectors, flexible UART/SPI mux)	Existing Dev Hardware (Open SDK / BSD-3)	**Satisfies L3-OBC-133, L3-OBC-150, L3-OBC-158, L3-OBC-167, L4-OBC-244–246, and L4-OBC-261–262 (ADR-001, ADR-002):** Provides a 520 KB SRAM frame buffer and dedicated compute pipeline for Chloropyhll Index calculation (L4-OBC-244) and S-Band SPI streaming (L4-OBC-200) without starving Main Computer real-time control loops, while drawing 0 W during Standby via EPS power gating (L4-OBC-261).
Real-Time Operating System (RTOS) Kernel	**FreeRTOS** (Preemptive priority-based microkernel)	MIT Open-Source License (v10.5+)	**Satisfies CON-02 (Open-Source), L3-OBC-139, L4-OBC-222, L4-OBC-247, L4-OBC-248, and L4-OBC-253:** Provides deterministic preemptive task scheduling (<100 ms fault alert preemption), static task/queue allocation (L4-OBC-248), and mutex primitives (L4-OBC-253) with zero licensing cost, resolving LEOPARDSat-1's monolithic state-machine maintenance bottleneck.
Core Flight Software Framework	**NASA F' (F Prime)** (Component-driven architecture with Svc::TlmChan, Svc::CmdDispatcher, Svc::Health)	Apache 2.0 Open-Source License (v3.4+)	**Satisfies CON-02, CON-03, CON-05, L3-OBC-131–132, L4-OBC-194–195, L4-OBC-248, and L4-OBC-250:** Scored highest in our framework trade study (4.60/5.00). Enforces static memory allocation (L4-OBC-248), typed message-passing ports, out-of-the-box command dispatching/telemetry channelization, and flight-proven Svc::Health watchdog wrapping (L4-OBC-250).

Layer	Technology Choice	License / Version	Justification
Flight Software Language & Build Toolchain	**Modern C++ (C++17)** compiled with **ARM GNU Toolchain (arm-none-eabi-gcc) & CMake**	GPLv3 with GCC Runtime Exception / BSD (Open-Source)	**Satisfies CON-02, CON-05, L4-OBC-235, and L4-OBC-248:** Matches CubeCats student skillsets while providing compile-time type safety, zero-cost abstractions, deterministic constexpr static buffer sizing, and zero vendor lock-in.
Telecommand & Downlink Cryptography	**ASCON-AEAD128** (Lightweight Authenticated Encryption with Associated Data)	CC0 / Public Domain Open-Source	**Satisfies CON-06, CON-08, L2-MIS-079, L3-OBC-129, L3-OBC-130, and L4-OBC-185–189:** Selected in our Cybersecurity Trade Study over Grain-128 and AES-GCM due to its compact 1,090-byte code size, 40-byte RAM footprint, interruptibility on Cortex-M4, ~2,547-cycle base execution speed, and integrated 128-bit authentication tag.
Non-Volatile Storage Architecture	**SPI MRAM** (State/Parameter Store) + **SPI NAND Flash** (Mass Image/Log Store with CRC32 & Bad-Block Remapping)	Hardware / Custom C++ Driver (Storage_SpaceManager)	**Satisfies L3-OBC-144–148, L4-OBC-212, L4-OBC-224–230, and L4-OBC-264:** SPI MRAM provides unlimited write endurance and radiation tolerance for 30-second state persistence (L4-OBC-212) and parameter tables (L4-OBC-264), while SPI NAND Flash retains >=7 multi-band image sets with CRC verification (L4-OBC-224) until Ground Station purge confirmation (L4-OBC-225).
Unit Testing & SITL Verification Suite	**GoogleTest (GTest) + F' STest + Python 3 Hardware-Mocked SITL Harness + GitHub Actions CI**	BSD-3 / Apache 2.0 Open-Source	**Satisfies CON-02, CON-05, and L4-OBC-256:** Directly addresses the LEOPARDSat-1 lesson learned (lack of off-hardware software tests) by mocking CAN, UART, SPI, and GPIO HAL interfaces so developers verify >=80% statement coverage on desktop machines without physical hardware access.
Ground Station Database & Dashboard Stack	**Python 3 F' GDS + SQLite 3 (SQLAlchemy ORM) + React TypeScript Dashboard**	Public Domain / MIT Open-Source	**Satisfies CON-02, L3-OBC-132, L3-OBC-135, and L4-OBC-202 (Tasks 022–026):** Provides a zero-configuration relational store (habsat_gds.db) on the Ground Station PC for archiving downlinked pwr_data, adcs_data, and imgr_data packets and serving validated JSON schemas to operator consoles.

Known Constraints

Version	Date	Change Description	Author
CON-01	2026-10-28	**Existing Club Hardware & Budget Limit:** Rules out commercial rad-hard SBCs or custom carrier PCBs. Forces deployment on EnduroSat OBC Type I (STM32F407VG) and RP2350 Data Processor (ADR-001, L3-OBC-167), keeping hardware capped $0.	Caden O'Leary
CON-02	2026-10-28	**Open-Source Tooling Mandate:** Rules out commercial RTOS and paid IDE compilers. Forces selection of FreeRTOS, NASA F', ARM GCC/CMake, GoogleTest, Python 3, and SQLite across flight and ground software stack.	Elmoataz Abdellah
CON-03	2026-10-28	**Aerospace Industry PDR:** Rules out ad-hoc coding and unverified state transitions. Forces formal L0–L4 requirement decomposition, ADRs 001–004, 80-item FMEA/Edge-Case catalog, and requirement-to-test traceability.	Matthew Goldsberry
CON-04	2026-10-28	**UC & CubeCats Public Representation:** Rules out unmoderated open repeater transmissions. Forces strict packet validation, 64-kb digipeater ring buffer (L4-OBC-207), max duty cycle (L4-OBC-208), and immediate RF inhibit capability (0x07, L3-OBC-171).	Seunghun Lee
CON-05	2026-10-28	**Aerospace Software Engineering Practices:** Rules out unreviewed code merges or relying solely on manual bench testing. Forces Git branch protection, automated CI with >=80% coverage off-target unit tests (L4-OBC-256), and standardized umbilical ports (L3-OBC-172).	Kristen Griessman
CON-06	2026-10-28	**U.S. Export Control (ITAR / EAR):** Rules out export-restricted cryptographic modules or committing keys to public repos. Forces use of a public ASCON-AEAD128 cipher with flight keys in write-protected onboard NVM (L4-OBC-188).	Caden O'Leary
CON-07	2026-10-28	**FCC & NOAA Spectrum / Remote Sensing Compliance:** Rules out immediate RF transmission or unauthorized imaging resolution. Forces 30-min deploy / 45-min RF inhibit timers in MRAM (L3-OBC-169, L4-OBC-212), RF kill switch (0x07, L3-OBC-171), and GSD check (L4-OBC-237).	Elmoataz Abdellah
CON-08	2026-10-28	**Downlink & Telecommand Encryption:** Rules out unauthenticated commands or unencrypted private downlinks. Forces ASCON-AEAD128 encryption and 128-bit MAC tags on non-amateur commands and downlinks (L3-OBC-129, L4-OBC-185–189).	Kristen Griessman
CON-09	2026-10-28	**Critical Command Multi-Step Validation:** Rules out single-packet execution of destructive commands (0x05, 0x06, 0x07, 0x09, 0x14). Forces secondary confirmation check in command dispatcher (L4-OBC-193, L4-OBC-225).	Seunghun Lee
CON-10	2026-10-28	**Subsystem Fault Isolation:** Rules out daisy-chaining subsystems on unprotected serial lines. Forces multi-drop CAN 2.0B with 3x reset (L4-OBC-252), EPS power-rail isolation (L4-OBC-263), dynamic UART gating, and EPS-to-UHF failover (ADR-004).	Matthew Goldsberry

Traceability Matrix

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
L0-PRJ-001	Science Objective: Demonstrate collection of multispectral images of HAB dynamics on Lake Erie from LEO.	D0: Multispectral Imager, Truth Imager, Imager Adapter, Main Computer, Data Processor	Designed	TC-001
L0-PRJ-002	Technology Objective: Demonstrate low-cost, drone-rated multispectral imager for remote sensing from space.	D0: Payload Stack (Multispectral Imager, Imager Adapter), Avionics Stack	Designed	TC-002
L0-PRJ-004	External Stakeholder Requirements: Comply with all external stakeholder requirements.	D0/D1/D2: Complete HABSat-1 Avionics, Comms, and Payload Architecture	Designed	TC-004
L1-SCI-008	Multispectral Imaging Data: Cover 620 nm, 650 nm, 680 nm, and 708 nm wavelengths with 10 nm FWHM bandwidth.	D0: Multispectral Imager, Imager Adapter; D1: CMD_IMG_TRIGGER_BURST (0x31); D2: MultiImg_FilterHoming (EC-57)	Designed	TC-008
L1-SCI-009	Imaging Ground Resolution: Multispectral images shall have a GSD <= 90 meters per pixel.	D0: Main Computer, ADCS Computer; D1: Pass Execution Sequencer Pre-Pass Check	Designed	TC-009
L1-SCI-010	Imaging Georeference: Each multispectral image shall have a coordinate reference locating its geographical center.	D0: Main Computer, ADCS Computer, GPS Receiver; D1: Image Georeference Metadata Header	Designed	TC-010
L1-SCI-012	Mission Length: Collect multispectral imagery of Lake Erie over at least one year.	D0: Avionics & Power Stack; D1: EnduroSat OBC Type I & FreeRTOS Static Architecture	Designed	TC-012
L1-MIS-015	Flight Element: Use a single prototype CubeSat to collect and downlink all multispectral imagery from orbit.	D0: HABSat-1 Integrated Spacecraft Architecture	Designed	TC-015
L1-PRJ-018	Amateur Radio Support: Support amateur radio community via repeater/beacon operations.	D0: Main Computer, Telemetry/Command Radio (UHF); D1: RADIO_BEACON_TX (0xA1), RADIO_DIGIPEAT_STAGING (0xA3)	Designed	TC-018
L1-PRJ-019	Communication Regulation Compliance: Comply with all US FCC regulations for radio frequencies.	D0: Telemetry/Command Radio, Main Downlink Radio; D1/D2: 45-min RF Inhibit & Enter RF Inhibit (0x07)	Designed	TC-019
L1-PRJ-020	Remote Sensing Regulation Compliance: Comply with all US NOAA remote sensing regulations.	D0: Main Computer, Payload Stack; D1/D2: ASCON-AEAD128 Downlink Encryption & GSD Check	Designed	TC-020
L1-PRJ-021	Export Control Compliance: Comply with all US EAR and ITAR export control and research security regulations.	D1/D2: Public-Domain ASCON-AEAD128 Cipher & Write-Protected NVM Key Storage	Designed	TC-021

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
L2-SYS-026	Form Factor: The CubeSat shall have a 3U form-factor.	D0: Structures & Avionics PC/104 Stack	Designed	TC-026
L2-SYS-027	Total Mass: The CubeSat shall have a mass <= 6000 grams.	D0: Spacecraft Mass Budget & Avionics Stack	Designed	TC-027
L2-SYS-038	Imaging Pointing: Actively point the multispectral imager to capture targeted area during an imaging pass.	D0: Main Computer, ADCS Computer, Attitude Actuators; D1: CMD_ADCS_SET_MODE (0x02)	Designed	TC-038
L2-SYS-040	Attitude Determination: Determine orientation in LVLH frame during capture of each image.	D0: ADCS Computer, Attitude Sensors; D2: ACDS_MatrixGuard (EC-01), SunSensor_Filter (EC-05)	Designed	TC-040
L2-SYS-041	Position Determination: Determine position in ECEF frame during capture of each image.	D0: ADCS Computer, GPS Receiver, GPS Antenna; D2: ACDS_NavFallback (EC-04), GPS_BoundsCheck (EC-13)	Designed	TC-041
L2-SYS-042	Image Data Storage: Store at least 7 multispectral images onboard at a given time.	D0: Data Storage (SPI NAND Flash); D1: DS_CHECK_IMAGE_INDEX (0x0200); D2: Storage_SpaceManager (EC-69)	Designed	TC-042
L2-SYS-043	Telemetry Collection: Collect health/status information of onboard systems for later downlink.	D0: Main Computer, Data Storage; D1: F' Svc::TlmChan, DS_WRITE_LOG (0x0100)	Designed	TC-043
L2-SYS-044	System Lifetime: Designed to operate for at least 2 years.	D1: EnduroSat OBC Type I, SPI MRAM, Wear-Leveled NAND Flash	Designed	TC-044
L2-SYS-046	Energy Balance: Maintain a positive orbit-average energy balance in all nominal operational modes.	D0: Power Computer, Solar Panels, Battery Charging; D1: CMD_PWR_RAIL_ON/OFF (0x21/0x22); D2: MPPT_ControlLoop (EC-26)	Designed	TC-046
L2-SYS-048	Safe Mode: Include a Safe Mode keeping the satellite power positive, thermally benign, and commandable.	D0: Main Computer, Power Computer; D2: Power_SoCSupervisor (EC-21), Safe Mode FSM	Designed	TC-048
L2-SYS-049	Commandability: Respond to and execute all valid ground station commands received.	D0: Main Computer, Telemetry/Command Radio (UHF); D1: F' Svc::CmdDispatcher; D2: GS Command API (0x00–0x16)	Designed	TC-049
L2-SYS-050	Internal Temperature: All internal elements operational over -40 deg C to 60 deg C.	D1: EnduroSat OBC Type I Hardware; D2: Charge_ThermalCutoff (EC-25), Downlink_ThermalControl (EC-37), TruthImager_ThermalProtect (EC-56)	Designed	TC-050
L2-SYS-053	Testing Interfaces: Include communication and power interfaces facilitating external testing when integrated.	D1: EnduroSat JP2/JP14 Umbilical Serial & SWD Programming Header	Designed	TC-053

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
L2-SYS-059	Radiation Survivability: Mitigate radiation effects encountered in mission orbit.	D1: Shielding, SPI MRAM, Golden Image Bootloader; D2: MainComp_BootLoader (EC-67), Storage_BadBlockManage (EC-71)	Designed	TC-059
L2-SYS-060	Heartbeat: Transmit a periodic beacon containing basic health telemetry when not actively downlinking.	D0: Main Computer, Telemetry/Command Radio, UHF Antenna; D1: RADIO_BEACON_TX (0xA1)	Designed	TC-060
L2-SYS-061	Downlink Capacity: Capable of downlinking at least 7 multispectral images per day.	D0: Data Processor, Main Downlink Radio (S-Band) (ADR-002); D1: CMD_DP_TRIGGER_DOWNLINK (0x12)	Designed	TC-061
L2-MIS-071	Deployment Wait Time: Do not deploy any deployables until >= 30 minutes have elapsed from dispenser exit.	D0: Main Computer, Deployment Mechanisms; D1: DS_READ_FLAGS (0x0010); D2: Deploy_SafetyTimer (EC-78)	Designed	TC-071
L2-MIS-072	Transmission Wait Time: Do not transmit any signals until >= 45 minutes have elapsed from dispenser exit.	D0: Main Computer, UHF & S-Band Radios; D1: DS_READ_FLAGS (0x0010), Post-Deployment Timer	Designed	TC-072
L2-MIS-073	Independent Startup: Independently perform all startup activities (init, detumbling, deployables) within 1 week.	D0: Main Computer, ADCS, EPS, Deployment Mechanisms; D1: Initialization & Detumble FSM (0x02, 0x41, 0x54)	Designed	TC-073
L2-MIS-074	Imaging Criteria: Attempt to image a target when the entire feature can be imaged at GSD < 90 m/pixel.	D0: Main Computer, ADCS Computer; D1: Pass Execution Sequencer Pre-Pass Check	Designed	TC-074
L2-MIS-075	Image Downlink: Downlink multispectral images from storage upon command by Ground Station.	D0: Main Computer, Data Processor, Main Downlink Radio; D2: Send Specified Images (0x11), Send Image via UHF (0x13)	Designed	TC-075
L2-MIS-078	Transmission Shutdown: Disable and reenable transmissions from any onboard transmitter via ground command.	D0: Main Computer, UHF & S-Band Radios; D2: Enter RF Inhibit (0x07), Exit RF Inhibit (0x08)	Designed	TC-078
L2-MIS-079	Command Encryption: All non-amateur radio commands to the CubeSat shall be encrypted.	D0: Main Computer, UHF Radio; D1/D2: ASCON-AEAD128 Cipher & Radio_SecurityHandler (EC-35)	Designed	TC-079
L2-MIS-080	Amateur Repeater: Act as an amateur radio repeater when in nominal mode and not imaging/downlinking.	D0: Main Computer, UHF Radio; D1: RADIO_DIGIPEAT_STAGING (0xA3); D2: Repeat Message (0x02), Radio_Arbiter (EC-34)	Designed	TC-080
L2-MIS-081	Semi-Autonomous Operation: Autonomously execute nominal mission sequence without real-time ground contact.	D0: Main Computer; D1: Pass Execution Sequencer & 10-Mode FSM	Designed	TC-081

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
L2-MIS-082	Command ACK & Rejection Reporting: Transmit execution confirmation or rejection status with error reason for every telecommand.	D0: Main Computer, UHF Radio; D1: RADIO_SEND_DOWNLINK (0xA2); D2: DL-12 Command ACK	Designed	TC-082
L2-MIS-083	Time-Tagged Scheduled Command Execution: Receive, queue, and autonomously execute time-tagged telecommands at future epoch timestamps.	D0: Main Computer; D1: Static Schedule Queue (N <= 16) & Epoch Timer Dispatcher	Designed	TC-083
L2-MIS-084	Verified Handshake Image Purging: Retain captured image files in NVM until explicit purge command from Ground Station.	D0: Main Computer, Data Storage, Imager Adapter; D1: CMD_IMG_PURGE_BUFFER (0x33); D2: Purge Stored Image (0x14)	Designed	TC-084
L3-OBC-129	Command Decryption: Decrypt encrypted telecommand packets received from Comms Subsystem prior to validation.	D0: Main Computer, Telemetry/Command Radio (UHF); D2: Radio_SecurityHandler (EC-35), ASCON-AEAD128	Designed	TC-129
L3-OBC-130	Command Authentication: Verify cryptographic authenticity of all non-amateur telecommand packets.	D0: Main Computer, Telemetry/Command Radio (UHF); D2: Radio_SecurityHandler (EC-35)	Designed	TC-130
L3-OBC-131	Command Validation: Validate packet structure, parameter bounds, and operational mode compatibility before execution.	D0: Main Computer; D1: F' Svc::CmdDispatcher; D2: Radio_BufferManager (EC-33), SecPayload_ScriptParser (EC-63)	Designed	TC-131
L3-OBC-132	Command Status Reporting: Transmit execution confirmation or rejection status code to Ground Station for every telecommand.	D0: Main Computer, Telemetry/Command Radio; D1: RADIO_SEND_DOWNLINK (0xA2); D2: DL-12 ACK	Designed	TC-132
L3-OBC-133	Primary Image Downlink Operation: Retrieve, package, and stream specified image data to S-Band radio upon GS command.	D0: Main Computer, Data Processor, Main Downlink Radio (ADR-002); D1: CMD_DP_TRIGGER_DOWNLINK (0x12); D2: Downlink_StreamManager (EC-38), Downlink_BlockRequester (EC-39)	Designed	TC-133
L3-OBC-134	Backup Image Downlink Operation: Route compressed image previews or metadata packets to UHF radio upon GS command.	D0: Main Computer, Telemetry/Command Radio; D1: RADIO_SEND_DOWNLINK (0xA2); D2: Send Image via UHF (0x13), DL-06	Designed	TC-134
L3-OBC-135	Telemetry Downlink Operation: Transmit most recent full system telemetry data to Comms Subsystem upon GS command.	D0: Main Computer, Telemetry/Command Radio; D1: RADIO_SEND_DOWNLINK (0xA2), Svc::TlmChan; D2: Transmit Telemetry (0x01), Radio_TxController (EC-36)	Designed	TC-135

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
L3-OBC-136	Error Log Downlink Operation: Transmit recorded historical fault and error logs to Comms Subsystem upon GS command.	D0: Main Computer, Data Storage, Telemetry/Command Radio; D1: DS_WRITE_LOG (0x0100); D2: Send Errors (0x0B), DL-11	Designed	TC-136
L3-OBC-137	Subsystem Mode Configuration: Configure operational state of all onboard subsystems according to active operational mode.	D0: Main Computer, FreeRTOS Scheduling; D1: CMD_ADCS_SET_MODE (0x02), CMD_PWR_RAIL_ON/OFF (0x21/0x22)	Designed	TC-137
L3-OBC-138	Subsystem Monitoring: Collect and analyze subsystem telemetry data from all onboard subsystems every 60 seconds.	D0: Main Computer; D1: CMD_ADCS_GET_STATUS (0x01), CMD_PWR_GET_VITALS (0x20), CMD_IMG_GET_STATUS (0x32); D2: Power_Parser (EC-23), GPS_HardwareMonitor (EC-16)	In progress	TC-138
L3-OBC-139	Asynchronous Subsystem Fault Handling: Process asynchronous fault alerts from onboard subsystems immediately upon receipt.	D0: Main Computer; D1: Async CAN Alert Handlers (0x44, 0x51, 0x63); D2: Power_SoCSupervisor (EC-21)	Designed	TC-139
L3-OBC-140	Heartbeat Operation: Construct heartbeat packet and send to Comms Subsystem immediately after telemetry collection.	D0: Main Computer, Telemetry/Command Radio, UHF Antenna; D1: RADIO_BEACON_TX (0xA1)	Designed	TC-140
L3-OBC-141	System Timekeeping: Maintain a system clock and spacecraft epoch time.	D0: Main Computer; D1: RTC & Monotonic Tick Counter; D2: GPS_TimeKeeper (EC-14), Deploy_SafetyTimer (EC-78)	Designed	TC-141
L3-OBC-142	Primary GPS Time Synchronization: Synchronize spacecraft epoch time using GPS time readings from ADCS Subsystem.	D0: Main Computer, ADCS Computer, GPS Receiver; D2: GPS_TimeKeeper (EC-14)	Designed	TC-142
L3-OBC-143	Fallback Ground Time Synchronization: Synchronize spacecraft epoch time using Ground Station telecommands when GPS unavailable.	D0: Main Computer, Telemetry/Command Radio; D2: Sync Time (0x16)	Designed	TC-143
L3-OBC-144	Data Storage: Provide at least 8 GB (baseline target) of onboard data storage.	D0: Data Storage (SPI NAND Flash); D2: Storage_SpaceManager (EC-69)	Designed	TC-144
L3-OBC-145	State Non-Volatile Storage: Provide radiation-tolerant NVM (MRAM) for critical system states, mission phase flags, and inhibit timers.	D0: Data Storage (SPI MRAM); D1: DS_READ_FLAGS / DS_WRITE_FLAG (0x0010); D2: Peer Review Action #3 (MRAM Block)	Designed	TC-145
L3-OBC-146	Storage Management: Store, compress, decompress, and delete data in storage.	D0: Main Computer, Data Processor, Data Storage; D1: SPI Storage Driver; D2: Storage_SpaceManager (EC-69),	In progress	TC-146

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
		Storage_FileLockArbiter (EC-70), Storage_BadBlockManage (EC-71)		
L3-OBC-147	Verified Image Storage Purging: Retain captured image files in NVM until explicit Ground Station command is received.	D0: Main Computer, Data Storage, Imager Adapter; D1: CMD_IMG_PURGE_BUFFER (0x33); D2: Purge Stored Image (0x14)	Designed	TC-147
L3-OBC-148	Data Storage Telemetry: Collect and analyze status and health information from onboard data storage devices.	D0: Main Computer, Data Storage; D2: Storage_SpaceManager (EC-69), Storage_BusRecovery (EC-72)	In progress	TC-148
L3-OBC-149	Main Computer Telemetry: Collect and analyze Main Computer status and health information.	D0: Main Computer; D2: MainComp_MemManager (EC-65), MainComp_Watchdog (EC-66), MainComp_CANSupervisor (EC-68)	Designed	TC-149
L3-OBC-150	Data Processor Telemetry: Collect and analyze health and processing status telemetry from the Data Processor.	D0: Main Computer, Data Processor; D1: MSG_DP_ACK (0x70), MSG_DP_DOWNLINK_STATUS (0x73); D2: Processor_DSPMonitor (EC-76)	In progress	TC-150
L3-OBC-151	TLE Forwarding: Forward TLEs, target coordinates, and flags to ADCS Subsystem upon receipt from Ground Station.	D0: Main Computer, ADCS Computer; D1: CMD_ADCS_LOAD_TLE (0x03), CMD_ADCS_SET_TARGET_LAT/LON/ALT (0x04–0x06); D2: Uplink Orbit TLE Set (0x12), Send Target Coordinates (0x15)	Designed	TC-151
L3-OBC-152	Image Constraint Verification: Verify all imaging constraints are met prior to imaging a target.	D0: Main Computer; D1: Pass Execution Sequencer Pre-Pass Check	Designed	TC-152
L3-OBC-153	Attitude Mode Commanding: Command ADCS Subsystem to transition between operational attitude modes based on schedule.	D0: Main Computer, ADCS Computer, Attitude Actuators; D1: CMD_ADCS_SET_MODE (0x02); D2: SBand_PointPredictor (EC-46)	Designed	TC-153
L3-OBC-154	Imaging Execution: Command Payload Subsystem imagers to capture multispectral images.	D0: Main Computer, Imager Adapter, Truth Imager, Multispectral Imager; D1: CMD_IMG_TRIGGER_BURST (0x31); D2: TruthImager_StateSequencer (EC-54)	Designed	TC-154
L3-OBC-155	Amateur Repeater Operation: Send received amateur radio transmission back to Comms Subsystem link to be rebroadcasted.	D0: Main Computer, Telemetry/Command Radio, UHF Antenna; D1: RADIO_DIGIPEAT_STAGING (0xA3); D2: Repeat Message (0x02), Radio_Arbiter (EC-34)	Designed	TC-155
L3-OBC-156	OBC Reset Handling: Reset itself and all other systems to pre-launch state when onboard deployment switches are cycled.	D0: Main Computer, Data Storage; D1: DS_WRITE_FLAG (0x0010), Pre-Launch Reset Handler	Designed	TC-156

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
L3-OBC-157	OBC Standby Power Consumption: Do not exceed 0.45 W (baseline target) during Standby Mode.	D0: Main Computer, Power Computer; D1: EnduroSat OBC Type I (104–123 mA) & CMD_PWR_RAIL_OFF (0x22)	Designed	TC-157
L3-OBC-158	OBC Peak Power Consumption: Do not exceed 2.50 W (baseline target) during active image processing.	D0: Main Computer, Data Processor, Power Computer; D1: EnduroSat OBC Type I + RP2350 Power Budget	Designed	TC-158
L3-OBC-159	Software Fault Recovery: Detect and recover from onboard software exceptions, task hangs, and memory corruption events.	D0: Main Computer; D1: F' Svc::Health Watchdog Wrapper; D2: MainComp_MemManager (EC-65), MainComp_Watchdog (EC-66), MainComp_BootLoader (EC-67)	Designed	TC-159
L3-OBC-160	Communication Bus Fault Recovery: Detect and isolate communication faults on internal data buses without compromising unaffected subsystems.	D0: Main Computer, Power Computer; D2: MainComp_CANSupervisor (EC-68), Storage_BusRecovery (EC-72), Peer Review Action #4 (UART Watchdog)	Designed	TC-160
L3-OBC-161	Safe Mode Transition: Transition spacecraft to Safe Mode autonomously upon unrecoverable faults or low-battery alerts, or upon GS command.	D0: Main Computer, Power Computer; D1: MSG_PWR_LOW_BATTERY_ALERT (0x51); D2: Power_SoCSupervisor (EC-21)	Designed	TC-161
L3-OBC-162	Safe Mode Exit: Transition spacecraft out of Safe Mode back to nominal modes only upon authenticated GS command.	D0: Main Computer, Telemetry/Command Radio; D2: Exit Safe Mode (0x03)	Designed	TC-162
L3-OBC-163	OBC Radiation Shielding: Include radiation shielding for Main Computer, Data Processor, and Data Storage components.	D0: Avionics Stack Enclosure & EnduroSat OBC Type I Hardware	Designed	TC-163
L3-OBC-164	OBC Mass Limit: Do not exceed 250 grams (baseline target) in mass.	D0: Avionics Stack Mass Allocation	Designed	TC-164
L3-OBC-165	OBC Volume: Fit within 95 mm x 95 mm x 25 mm PC/104 envelope.	D0/D1: EnduroSat OBC Type I PC/104 Mechanical Form Factor	Designed	TC-165
L3-OBC-166	Radiation-Resistant Reboot: Overwrite flight software from radiation-resistant memory device during reboot.	D0: Main Computer, Data Storage; D2: MainComp_BootLoader (EC-67)	Designed	TC-166
L3-OBC-167	On-Board Chlorophyll Index Processing: Include dedicated Data Processor executing Chlorophyll Index conversion on captured imagery.	D0: Data Processor (ADR-001), Data Storage; D1: CMD_DP_START_PROCESSING (0x10); D2: Processor_MathGuard (EC-73), Processor_DynamicScheduler (EC-74)	In progress	TC-167

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
L3-OBC-168	Schedule Telecommand Execution: Store time-tagged telecommands in schedule queue and execute at specified spacecraft epoch.	D0: Main Computer; D1: Static Schedule Queue (N <= 16)	Designed	TC-168
L3-OBC-169	Post-Deployment Inhibit Enforcement: Enforce minimum 30-min deployable wait timer and 45-min RF transmission wait timer.	D0: Main Computer, Data Storage, Deployment Mechanisms; D1: DS_READ_FLAGS (0x0010); D2: Deploy_SafetyTimer (EC-78)	Designed	TC-169
L3-OBC-170	Autonomous Startup Sequencing: Autonomously coordinate deployable release and initial detumbling within 1 week of dispenser exit.	D0: Main Computer, ADCS, EPS, Deployment Mechanisms; D1: PIN_DEP_*, CMD_ADCS_SET_MODE (0x01), MSG_ADCS_DETUMBLE_COMPLETE (0x41), MSG_PWR_DEPLOY_CONFIRM (0x54)	Designed	TC-170
L3-OBC-171	Commanded RF Transmission Inhibit: Disable and reenable all onboard RF transmissions upon valid GS shutdown command.	D0: Main Computer, UHF & S-Band Radios; D2: Enter RF Inhibit (0x07), Exit RF Inhibit (0x08)	Designed	TC-171
L3-OBC-172	Integrated Ground Diagnostic Interface: Provide external umbilical interface accessible when integrated for flashing and diagnostics.	D1: EnduroSat OBC Type I JP2/JP14 Umbilical Interface	Designed	TC-172
L3-OBC-173	Image Georeference Metadata Tagging: Associate synchronized ECEF position and LVLH attitude estimates with each captured image set.	D0: Main Computer, ADCS Computer; D1: Image Georeference Sidecar Header	Designed	TC-173
L3-OBC-174	Thermal Control Actuation: Enforce subsystem thermal operating inhibits based on collected temperature telemetry.	D0: Main Computer, EPS, Payload, Comms; D2: Charge_ThermalCutoff (EC-25), Downlink_ThermalControl (EC-37), TruthImager_ThermalProtect (EC-56)	Designed	TC-174
L3-OBC-175	On-Orbit Parameter Reconfiguration: Support updating configurable operational thresholds, timeouts, and coefficients via telecommand.	D0: Main Computer, Data Storage (MRAM); D1: Non-Volatile Parameter Table	Planned	TC-175
L3-OBC-176	On-Orbit Flight Software Reprogrammability: Support receiving, verifying, and installing updated flight software images on orbit.	D0: Main Computer, Data Storage; D1: Bootloader & Staging Partition (Task 002)	Planned	TC-176
L3-OBC-177 to L3-OBC-184	[WITHDRAWN] Draft requirements 177–184 retired during D1/D2 decomposition without renumbering.	Withdrawn (IDs permanently retired)	Designed	N/A (Withdrawn)
L4-OBC-185	Telecommand Decryption: Decrypt encrypted telecommand frames prior to command parsing.	D0: Main Computer, Telemetry/Command Radio; D2: Radio_SecurityHandler (EC-35), ASCON-AEAD128 (Task 024)	In progress	TC-185

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
L4-OBC-186	Telecommand Authentication: Verify cryptographic authentication tag of each non-amateur telecommand frame prior to execution.	D0: Main Computer, Telemetry/Command Radio; D2: Radio_SecurityHandler (EC-35), ASCON-AEAD128 (Task 024)	In progress	TC-186
L4-OBC-187	Telecommand Replay Protection: Reject non-amateur telecommand frames with duplicate or out-of-window sequence counter.	D0: Main Computer; D2: Radio_SecurityHandler (EC-35) Replay Window Counter	Designed	TC-187
L4-OBC-188	Cryptographic Key Isolation: Store cryptographic keys in write-protected NVM inaccessible to unauthenticated interfaces.	D0: Main Computer, Data Storage (MRAM/OTP Flash); D2: Protected Key Sector	Designed	TC-188
L4-OBC-189	Unauthenticated Frame Rejection: Discard frames failing decryption/authentication and increment security rejection counter.	D0: Main Computer; D2: Radio_SecurityHandler (EC-35)	Designed	TC-189
L4-OBC-190	Command Syntax Validation: Validate packet length, structure, and checksum of each received telecommand frame.	D0: Main Computer; D1: F' Svc::CmdDispatcher; D2: Radio_BufferManager (EC-33)	Designed	TC-190
L4-OBC-191	Command Parameter Bounds Checking: Verify all command arguments fall within defined operational boundary limits prior to dispatch.	D0: Main Computer; D2: Command Parameter Validator (EC-11, EC-59, EC-63)	Designed	TC-191
L4-OBC-192	Command Mode Authorization: Reject telecommands not authorized for execution in the active spacecraft operational mode.	D0: Main Computer; D1: Operational Mode State Machine Guard	Designed	TC-192
L4-OBC-193	Critical Command Safeguard: Require secondary confirmation check prior to executing reboot (0x06), power cycle (0x05), RF inhibit (0x07), Emergency Mode (0x09), or data purge (0x14).	D0: Main Computer; D2: Two-Step Confirmation State Machine (0x05, 0x06, 0x07, 0x09, 0x14)	Designed	TC-193
L4-OBC-194	Command Execution Acknowledgment: Generate and queue positive ACK packet for every telecommand that passes validation and executes.	D0: Main Computer, UHF Radio; D1: RADIO_SEND_DOWNLINK (0xA2); D2: DL-12 Command ACK	Designed	TC-194
L4-OBC-195	Command Rejection Reporting: Generate and queue negative NACK packet with error reason code for every rejected telecommand.	D0: Main Computer, UHF Radio; D1: RADIO_SEND_DOWNLINK (0xA2); D2: E1 Exception Flow	Designed	TC-195
L4-OBC-196	Command Outcome Logging: Record execution or rejection status of every received telecommand in non-volatile log storage.	D0: Main Computer, Data Storage; D1: DS_WRITE_LOG (0x0100)	Designed	TC-196

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
L4-OBC-197	Scheduled Command Queueing: Store valid telecommands tagged with future execution time in fixed-capacity schedule queue.	D0: Main Computer; D1: Static Schedule Queue (N <= 16)	Designed	TC-197
L4-OBC-198	Scheduled Command Dispatch: Autonomously dispatch queued time-tagged telecommands when spacecraft epoch clock reaches target time.	D0: Main Computer; D1: FreeRTOS Epoch Timer Dispatcher	Designed	TC-198
L4-OBC-199	Downlink Frame Segmentation: Segment image and telemetry files into sequential transfer frames with sequence numbers and CRCs.	D0: Main Computer, Data Processor; D1: File Chunking Module (Task 020); D2: Downlink_StreamManager (EC-38)	In progress	TC-199
L4-OBC-200	High-Rate Downlink Flow Control: Regulate data transfer to high-rate S-Band transmitter based on buffer status feedback.	D0: Data Processor, Main Downlink Radio (ADR-002); D1: MSG_DP_DOWNLINK_STATUS (0x73); D2: Downlink_StreamManager (EC-38)	Designed	TC-200
L4-OBC-201	Low-Rate Image Preview Routing: Route compressed image previews and metadata packets to UHF telemetry transceiver upon command.	D0: Main Computer, Telemetry/Command Radio; D1: RADIO_SEND_DOWNLINK (0xA2); D2: Send Image via UHF (0x13), DL-06	Designed	TC-201
L4-OBC-202	On-Demand Telemetry Packaging: Compile current subsystem health, historical logs, and fault flags into transfer frames upon request.	D0: Main Computer, Data Storage; D1: F' Svc::TlmChan, RADIO_SEND_DOWNLINK (0xA2); D2: Transmit Telemetry (0x01), Send Errors (0x0B)	Designed	TC-202
L4-OBC-203	Housekeeping Beacon Assembly: Assemble unencrypted beacon containing time, mode, battery mV/SoC, temperatures, boot count, and fault flags.	D0: Main Computer; D1: RADIO_BEACON_TX (0xA1) Packet Builder	Designed	TC-203
L4-OBC-204	Housekeeping Beacon Cadence: Dispatch housekeeping beacon packet at a fixed 49-second interval when RF enabled and no downlink active.	D0: Main Computer, Telemetry/Command Radio; D1: RADIO_BEACON_TX (0xA1) 49-s Timer	Designed	TC-204
L4-OBC-205	Beacon Low-Power Suppression: Automatically suspend housekeeping beacon when in Low-Power Mode or RF Inhibit active.	D0: Main Computer; D1: Mode State Machine & Enter Low-Power Mode (0x04) / Enter RF Inhibit (0x07)	Designed	TC-205
L4-OBC-206	Uplink Reception Priority: Prioritize incoming telecommand reception over outgoing beacon or repeater TX on half-duplex UHF channel.	D0: Main Computer, Telemetry/Command Radio; D2: Radio_Arbiter (EC-34)	Designed	TC-206

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
L4-OBC-207	Amateur Repeater Buffering: Stage valid incoming amateur repeater packets in dedicated circular ring buffer of >= 64 kilobits.	D0: Main Computer; D1: RADIO_DIGIPEAT_STAGING (0xA3) 64-kb Static Ring Buffer	Designed	TC-207
L4-OBC-208	Amateur Repeater Rate Limiting: Enforce maximum transmission duty cycle on rebroadcast amateur radio repeater packets.	D0: Main Computer, Telemetry/Command Radio; D1: Repeater Duty-Cycle Limiter	Designed	TC-208
L4-OBC-209	Operational Mode State Machine: Execute deterministic FSM across Pre-Launch, Initialization, Checkout, Standby, Imaging, Downlink, Repeater, Low-Power, Safe, and Emergency modes.	D0: Main Computer; D1: 10-Mode Finite State Machine	Designed	TC-209
L4-OBC-210	Subsystem Power Rail Commanding: Command Power Subsystem to enable or disable individual subsystem rails according to active mode.	D0: Main Computer, Power Computer; D1: CMD_PWR_RAIL_ON (0x21), CMD_PWR_RAIL_OFF (0x22), MSG_PWR_ACK (0x50)	Designed	TC-210
L4-OBC-211	Graceful Subsystem Shutdown: Issue shutdown notification and verify pending data writes before removing subsystem power rails.	D0: Main Computer, Data Storage, Subsystems; D1: DS_CHECK_IMAGE_INDEX (0x0200); D2: SecPayload_ShutdownHandler (EC-64)	Designed	TC-211
L4-OBC-212	Non-Volatile State Persistence: Record active mode, mission phase flags, boot counter, and cumulative inhibit timers in MRAM every 30 s.	D0: Main Computer, Data Storage (MRAM); D1: DS_WRITE_FLAG (0x0010); D2: Peer Review Action #3	Designed	TC-212
L4-OBC-213	Post-Deployment Inhibit Timers: Enforce 30-min cumulative uptime delay before deployables and 45-min delay before RF transmissions.	D0: Main Computer, Data Storage; D1: DS_READ_FLAGS (0x0010); D2: Deploy_SafetyTimer (EC-78)	Designed	TC-213
L4-OBC-214	Deployable Actuation Current Limiting: Limit release mechanism actuation current to <= 2.5 A during deployable release sequences.	D0: Main Computer, Deployment Mechanisms; D1: PIN_DEP_*; D2: Deploy_CurrentLimiter (EC-77)	Designed	TC-214
L4-OBC-215	Deployable Feedback Debouncing: Filter deployable status switch signals over 100 ms debounce window and retry up to 2 times.	D0: Main Computer, Deployment Mechanisms; D2: Deploy_Debouncer (EC-79), UHF_DeployVerifier (EC-43)	Designed	TC-215
L4-OBC-216	Pre-Launch Reset Switch Handling: Zero persisted post-deployment timers and clear mission phase flags in NVM when switches cycle.	D0: Main Computer, Data Storage; D1: DS_WRITE_FLAG (0x0010)	Designed	TC-216
L4-OBC-217	Monotonic Clock Maintenance: Maintain continuous monotonic tick counter and Real-Time Clock with drift rate <= 20 ppm.	D0: Main Computer (EnduroSat OBC Type I RTC); D2: GPS_TimeKeeper (EC-14)	Designed	TC-217

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
L4-OBC-218	Primary GPS Time Synchronization: Discipline spacecraft epoch clock using GPS time messages and 1PPS pulses from ADCS.	D0: Main Computer, ADCS Computer, GPS Receiver; D2: GPS_TimeKeeper (EC-14)	Designed	TC-218
L4-OBC-219	Fallback Ground Time Synchronization: Update spacecraft epoch clock upon valid Sync Time (0x16) telecommand when GPS unavailable.	D0: Main Computer, Telemetry/Command Radio; D2: Sync Time (0x16)	Designed	TC-219
L4-OBC-220	Subsystem Telemetry Polling Scheduler: Poll health and status telemetry from Power, ADCS, Comms, Thermal, and Payload every 60 s.	D0: Main Computer; D1: CMD_ADCS_GET_STATUS (0x01), CMD_PWR_GET_VITALS (0x20), CMD_IMG_GET_STATUS (0x32) (Tasks 008, 012)	In progress	TC-220
L4-OBC-221	Telemetry Threshold Evaluation: Compare collected telemetry against mode bounds and assert fault flag when exceeded.	D0: Main Computer; D1: Telemetry Fault Evaluator (Tasks 010, 014)	In progress	TC-221
L4-OBC-222	Asynchronous Fault Alert Handling: Preempt background tasks to process critical asynchronous fault alerts within 100 ms.	D0: Main Computer; D1: FreeRTOS High-Priority ISR/Task (0x44, 0x51, 0x63)	Designed	TC-222
L4-OBC-223	Lossless Image Compression: Compress captured multispectral and truth image files losslessly prior to downlink packaging.	D0: Data Processor; D2: Processor_DynamicScheduler (EC-74)	In progress	TC-223
L4-OBC-224	Stored File Integrity Verification: Compute and store a CRC value for every file written to NVM mass storage and verify upon readback.	D0: Main Computer, Data Processor, Data Storage; D1: SPI File CRC32 Verifier	Designed	TC-224
L4-OBC-225	Ground-Verified Image Deletion: Delete stored image files and clear camera buffers only upon authenticated Purge Stored Image (0x14) command.	D0: Main Computer, Data Storage, Imager Adapter; D1: CMD_IMG_PURGE_BUFFER (0x33); D2: Purge Stored Image (0x14), Design Rationale #6	Designed	TC-225
L4-OBC-226	Storage Capacity Telemetry: Measure and report available free capacity on NVM mass storage every 60 seconds.	D0: Main Computer, Data Storage; D2: Storage_SpaceManager (EC-69) (Task 021)	In progress	TC-226
L4-OBC-227	Flash Bad-Block Remapping: Automatically remap write operations away from defective NVM blocks and increment bad-block counter.	D0: Data Storage; D2: Storage_BadBlockManage (EC-71)	Designed	TC-227
L4-OBC-228	Storage Wear-Leveling Telemetry: Distribute writes across NVM blocks and report wear-leveling status in storage telemetry.	D0: Data Storage; D1: Circular Buffer & Wear-Leveling Manager (Task 019)	In progress	TC-228

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
L4-OBC-229	High Storage Utilization Warning: Assert storage warning telemetry flag when NVM mass storage exceeds 85% capacity.	D0: Main Computer, Data Storage; D2: Storage_SpaceManager (EC-69)	Designed	TC-229
L4-OBC-230	Telemetry Log FIFO Rollover: Overwrite oldest unflagged telemetry logs in FIFO order at 99% capacity while preserving science images.	D0: Main Computer, Data Storage; D2: Storage_SpaceManager (EC-69)	Designed	TC-230
L4-OBC-231	Processor Temperature Telemetry: Sample and report internal temperatures of Main Computer and Data Processor every 60 seconds.	D0: Main Computer, Data Processor; D1: Internal Thermal ADC Readout (Task 004, Task 021)	Designed	TC-231
L4-OBC-232	Processor Voltage and Current Telemetry: Sample and report supply voltages and current draw of Main Computer and Data Processor every 60 s.	D0: Main Computer, Data Processor, Power Computer; D1: MSG_PWR_VITALS_DATA (0x53)	Designed	TC-232
L4-OBC-233	Processor Resource Utilization Telemetry: Record and report CPU load, RAM utilization, uptime, and reboot counters every 60 seconds.	D0: Main Computer, Data Processor; D1: F' Svc::Health & MainComp_MemManager (EC-65)	Designed	TC-233
L4-OBC-234	Ephemeris Target Forwarding: Forward updated TLEs (0x12) or target coordinates (0x15) to ADCS and request updated pass schedule (0x07).	D0: Main Computer, ADCS Computer; D1: CMD_ADCS_LOAD_TLE (0x03), CMD_ADCS_SET_TARGET_LAT/LON/ALT (0x04–0x06), CMD_ADCS_REQ_SCHEDULE (0x07) (Task 011)	Designed	TC-234
L4-OBC-235	Upcoming Pass Schedule Queue: Maintain fixed-capacity queue of up to 16 upcoming target imaging passes received from ADCS (0x43).	D0: Main Computer; D1: Pass Execution Sequencer PassRecord[16] Static Queue	Designed	TC-235
L4-OBC-236	Pass Schedule Conflict Resolution: Discard duplicate pass entries and resolve overlapping windows by higher priority or max elevation.	D0: Main Computer; D1: Pass Execution Sequencer Edge Case Handlers (Duplicate & Tie/Overlap Resolution)	Designed	TC-236
L4-OBC-237	Pre-Pass Constraint Verification: Verify battery SoC >= 20%, target GSD < 90 m/pixel, and zero active critical faults prior to pass.	D0: Main Computer, Power Computer, ADCS; D1: Pass Execution Sequencer Pre-Check	Designed	TC-237
L4-OBC-238	Pre-Pass Target Preparation: Enable Payload power rail (0x21), command ADCS target tracking (0x02), and start sensor warmup (0x30) at lead time.	D0: Main Computer, EPS, ADCS, Imager Adapter; D1: Pass Execution Sequencer Lead Phase	Designed	TC-238
L4-OBC-239	Synchronized Capture Triggering: Trigger capture (0x31) only after receiving both MSG_ADCS_TARGET_ACQUIRED (0x42) and MSG_IMG_WARMUP_READY (0x61).	D0: Main Computer, ADCS, Imager Adapter; D1: Pass Execution Sequencer Capture Gate	Designed	TC-239

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
L4-OBC-240	Post-Pass Standby Return: Power down Payload rail (0x22) and command ADCS to Standby/Minimal Drag (0x03) upon MSG_IMG_CAPTURE_COMPLETE (0x62).	D0: Main Computer, EPS, ADCS, Imager Adapter; D1: Pass Execution Sequencer Teardown Phase	Designed	TC-240
L4-OBC-241	Imaging Pass Fault Abort: Abort pass, cut Payload rail (0x22), and stop tracking (0x08) on warmup timeout, imager fault (0x63), lock timeout (0x44), or low battery (0x51).	D0: Main Computer, EPS, ADCS, Imager Adapter; D1: Pass Execution Sequencer Abort Handler	Designed	TC-241
L4-OBC-242	Ground Pass Preemption: Immediately preempt active imaging pass if incoming Ground Station telecommand session is detected.	D0: Main Computer, Telemetry/Command Radio, ADCS, EPS; D1: RADIO_CARRIER_DETECT Preemption Handler	Designed	TC-242
L4-OBC-243	Monotonic Pass Timeout Evaluation: Evaluate all in-pass phase timeouts using monotonic tick counter rather than wall-clock subtraction.	D0: Main Computer; D1: Pass Execution Sequencer Monotonic Clock Guard	Designed	TC-243
L4-OBC-244	Chloropyhll Index Computation: Command Data Processor (0x10) to ingest raw multispectral frames and compute Chloropyhll Index map.	D0: Main Computer, Data Processor (ADR-001); D1: CMD_DP_START_PROCESSING (0x10), MSG_DP_PROCESSING_DONE (0x72) (Task 003)	In progress	TC-244
L4-OBC-245	Raw Frame Transfer Retry: Verify checksum of each raw image block transferred from Payload and retry corrupted blocks up to 3 times.	D0: Imager Adapter, Data Processor; D2: Processor_FrameVerifier (EC-75)	Designed	TC-245
L4-OBC-246	Image Processing Arithmetic Guard: Detect divide-by-zero or overflow during CI computation and fall back to storing raw uncompressed image.	D0: Data Processor; D2: Processor_MathGuard (EC-73)	Designed	TC-246
L4-OBC-247	Preemptive Priority Task Scheduling: Execute tasks using preemptive priority-based FreeRTOS scheduling.	D0: Main Computer; D1: FreeRTOS Kernel Configuration	Designed	TC-247
L4-OBC-248	Static Memory Allocation: Allocate all tasks, queues, and buffers statically at initialization with zero dynamic heap allocation in flight.	D0: Main Computer; D1: NASA F' Static Ports & Buffers (Task 001), Design Rationale #5	In progress	TC-248
L4-OBC-249	Memory Utilization Protection: Suspend lowest-priority background tasks if runtime RAM utilization exceeds 95% of available memory.	D0: Main Computer; D2: MainComp_MemManager (EC-65)	Designed	TC-249
L4-OBC-250	Hardware Watchdog Reset: Strobe independent hardware watchdog timer during	D0: Main Computer; D1: F' Svc::Health Wrapper & Triple Watchdog Architecture; D2: MainComp_Watchdog (EC-66)	Designed	TC-250

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
	nominal execution that resets processor if execution hangs.			
L4-OBC-251	Golden Image Boot Recovery: Verify primary flight image CRC at boot and boot from write-protected fallback image after 3 failed boots.	D0: Main Computer; D2: MainComp_BootLoader (EC-67)	Designed	TC-251
L4-OBC-252	Internal Bus Fault Recovery: Retry failed internal bus transactions up to 3 times and automatically reset local bus controller on bus-off/timeout.	D0: Main Computer; D2: MainComp_CANSupervisor (EC-68), ACDS_BusParser (EC-03), Power_Parser (EC-23), Peer Review Action #4	Designed	TC-252
L4-OBC-253	Concurrent Storage Access Serialization: Serialize concurrent NVM read/write requests via mutex locking and buffer logs in RAM if busy/faulted.	D0: Main Computer, Data Processor, Data Storage; D2: Storage_FileLockArbiter (EC-70), Storage_BusRecovery (EC-72)	Designed	TC-253
L4-OBC-254	Autonomous Safe Mode Entry: Autonomously transition to Safe Mode and shed non-essential loads on unrecoverable fault, bus failure, or low battery.	D0: Main Computer, Power Computer; D2: Power_SoCSupervisor (EC-21)	Designed	TC-254
L4-OBC-255	Authenticated Safe Mode Exit: Clear latched Safe Mode fault flags and return to nominal modes only upon valid Ground Station Exit Safe Mode (0x03).	D0: Main Computer, Telemetry/Command Radio; D2: Exit Safe Mode (0x03)	Designed	TC-255
L4-OBC-256	Off-Target Hardware-Mocked Unit Testing: Pass automated off-target unit/integration tests with simulated hardware and >= 80% statement coverage.	D1: GTest / F' STest SITL Harness (Task 009)	In progress	TC-256
L4-OBC-257	Thermal Setpoint Regulation: Evaluate component temperature telemetry every 60 s and command heater rails or thermal operational inhibits.	D0: Main Computer, EPS, Payload, Comms; D2: Charge_ThermalCutoff (EC-25), Downlink_ThermalControl (EC-37), TruthImager_ThermalProtect (EC-56)	Designed	TC-257
L4-OBC-258	Power Subsystem Watchdog Heartbeat: Transmit periodic communication heartbeat to Power Subsystem within watchdog timeout window.	D0: Main Computer, Power Computer; D1: Watchdog Petting Sequence (Task 015)	In progress	TC-258
L4-OBC-259	Onboard Auxiliary Sensor Readout: Sample EnduroSat board-level accelerometer and magnetometer during telemetry collection.	D0: Main Computer (EnduroSat OBC Type I Integrated Sensors)	Planned	TC-259

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
L4-OBC-260	Image Georeference Sidecar Generation: Query synchronized ECEF position and LVLH attitude from ADCS upon capture completion and record in metadata header.	D0: Main Computer, ADCS Computer, Data Storage; D1: Georeference Metadata Tagging	Designed	TC-260
L4-OBC-261	Data Processor Power Gating: Keep Data Processor power rail disabled during Standby Mode, enabling only during image processing and S-Band downlink.	D0: Main Computer, Power Computer, Data Processor; D1: CMD_PWR_RAIL_ON/OFF (0x21/0x22, Rail 0x01)	Designed	TC-261
L4-OBC-262	Data Processor Execution Timeout: Enforce maximum execution timer on Data Processor operations, aborting (0x11) and cutting power (0x22) on timeout.	D0: Main Computer, Data Processor, Power Computer; D1: CMD_DP_ABORT (0x11), CMD_PWR_RAIL_OFF (0x22)	Designed	TC-262
L4-OBC-263	Faulted Bus Node Isolation: Sequentially command Power Subsystem to disable non-essential subsystem rails if shared bus stays locked after 3 resets.	D0: Main Computer, Power Computer; D1: CMD_PWR_RAIL_OFF (0x22); D2: MainComp_CANSupervisor (EC-68), Task 016	Designed	TC-263
L4-OBC-264	Non-Volatile Parameter Table Management: Store configurable thresholds, timeouts, and calibration coefficients in MRAM parameter table updatable via telecommand.	D0: Main Computer, Data Storage (MRAM); D1: MRAM Parameter Store	Planned	TC-264
L4-OBC-265	Chunked Software Image Staging: Receive segmented flight software update binaries over telecommand link, assemble in staging NVM, and report missing indices.	D0: Main Computer, Data Storage, Telemetry/Command Radio; D1: Software Staging Module (Task 002)	Planned	TC-265
L4-OBC-266	Staged Software Signature Verification: Verify cryptographic signature and CRC of staged software image prior to install and roll back after 24 h if unconfirmed.	D0: Main Computer, Data Storage; D1: Bootloader Signature Verifier (Task 002); D2: MainComp_BootLoader (EC-67)	Planned	TC-266
L4-OBC-267	Umbilical Diagnostic Command Interface: Provide external wired serial diagnostic interface supporting real-time telemetry, command injection, and pre-launch verification.	D0: Main Computer; D1: EnduroSat JP2/JP14 Diagnostic UART Interface	Designed	TC-267
L4-OBC-268	Umbilical Flight Software Flashing: Expose hardware programming lines through external umbilical to allow reflashing Main Computer and Data Processor when integrated.	D0: Main Computer, Data Processor; D1: External SWD / JTAG Umbilical Lines	Designed	TC-268
CON-01	Economic Constraint 1: Limited to already-purchased club hardware (EnduroSat OBC Type I) and budget.	D0/D1: EnduroSat OBC Type I Hardware Selection	Designed	TC-301

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
CON-02	Economic Constraint 2: Use open-source tooling (NASA F', FreeRTOS, GCC, Python, SQLite) and no proprietary software.	D1: Build-vs-Reuse Table & Open-Source Technology Stack	Designed	TC-302
CON-03	Professional Constraint 1: Undergo Preliminary Design Review (PDR) by aerospace industry professionals.	D0/D1/D2: Complete Design Baseline, ADRs 001–004, FMEA Risk Matrix, and Traceability Matrix	In progress	TC-303
CON-04	Professional Constraint 2: Actions and work reflect public opinion of University of Cincinnati and CubeCats.	D1/D2: Validated UHF Beacon (L4-OBC-203) & Rate-Limited Amateur Digipeater (L4-OBC-207–208)	Designed	TC-304
CON-05	Professional Constraint 3: Follow established aerospace software engineering practices (requirements, Git PRs, testing, docs, verification).	D1/D2: Stable L0–L4 IDs, Git PR Policy, GTest/F' SITL Suite (L4-OBC-256)	In progress	TC-305
CON-06	Legal Constraint 1: Comply with all US regulations relating to export control (ITAR/EAR) and research security (L1-PRJ-021).	D1/D2: Public-Domain ASCON-AEAD128 Cipher & Write-Protected NVM Key Isolation (L4-OBC-188)	Designed	TC-306
CON-07	Legal Constraint 2: Comply with all US regulations for utilization of radio frequencies (FCC) and remote sensing (NOAA) (L1-PRJ-019, L1-PRJ-020).	D1/D2: Post-Deployment Inhibit Timers (L4-OBC-213), Enter RF Inhibit (0x07, L3-OBC-171), GSD Check (L4-OBC-237)	Designed	TC-307
CON-08	Security Constraint 1: Encrypt downlink payload data and non-amateur commands (L2-MIS-079, L3-OBC-129, L4-OBC-185).	D1/D2: ASCON-AEAD128 Encryption on Commands (0x00–0x01, 0x03–0x16) and Downlinks (DL-05, DL-06, DL-10, DL-11)	In progress	TC-308
CON-09	Security Constraint 2: Require additional validation for critical spacecraft commands before execution (L4-OBC-193).	D1/D2: Critical Command Two-Step Confirmation Guard (L4-OBC-193) & Ground-Verified Image Purge (L4-OBC-225)	Designed	TC-309
CON-10	Security Constraint 3: Isolate faults within individual subsystems so one failure cannot compromise the OBC (L3-OBC-160, L4-OBC-263).	D0: ADR-004 EPS-to-Radio Failover; D1/D2: CAN 3x Reset (L4-OBC-252), EPS Node Isolation (L4-OBC-263), Dynamic UART Gating	Designed	TC-310

Record of Advisor Review

Faculty Advisor: Dr. Will Hawkins III, Department of Computer Science, University of Cincinnati (hawkinwh@ucmail.uc.edu)

Version	Date	Change Description	Author
Session 1	2026-09-15	Project Scope & Team Contract Approval: Reviewed Team Contract v1.0, Week 3 Constraints Essay (CON-01 to CON-10), LEOPARDSat-1 lessons learned, and initial CPU/RTOS trade study scope. Approved Team Contract via email sign-off. Emphasized addressing state-machine bottleneck early via formal RTOS study and structuring software for off-hardware unit testing without breadboard dependencies.	Dr. Will Hawkins III, Caden O'Leary, Seunghun Lee, Matthew Goldsberry, Elmoataz Abdellah
Session 2	2026-09-22	RTOS Selection, Failover Links & Command Security Review: Reviewed FSW Framework Trade Study (NASA F', cFS, FreeRTOS, Zephyr, RODOS, VxWorks), initial L4 OBC requirements, and D0 subsystem comms topology. Recommended adding direct EPS-to-UHF failover link (ADR-004), cryptographic encryption/authentication on non-amateur telecommands (ASCON-AEAD128), and ADCS GPS 1PPS time sync.	Dr. Will Hawkins III, Caden O'Leary, Matthew Goldsberry, Elmoataz Abdellah
Session 3	2026-10-13	D2 Detailed Design, Command Tagging & Traceability Review: Reviewed D1/D2 subsystem interface contracts, CAN/UART/SPI framing, edge-case fault tables (EC-01–EC-80), and traceability matrix. Added explicit source/destination command tagging across EPS and Main Computer frames, EPS watchdog heartbeat (L4-OBC-258), and bounded CAN/UART bus lockup recovery paths (3x retry, controller reset, power rail isolation).	Dr. Will Hawkins III, Caden O'Leary, Matthew Goldsberry, Elmoataz Abdellah, Kristen Griessman